

William Lu

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Education

University of Illinois Urbana-Champaign (UIUC)

B.S. in Mathematics & Computer Science

Sep 2024 - Expected Dec 2026

GPA: 4.0/4.0

University of Wisconsin Madison

B.S. in Computer Science (Transferred to UIUC in Sep 2024)

Sep 2023 - May 2024

GPA: 4.0/4.0

Relevant Course: Machine Learning, Applied Parallel Programming, Algorithm Design & Computation Models, Computer Systems Engineering, Data Structures, etc.

Experiences & Project

TikTok - Effect Core SDK Intern

AI Agent Engineer

San Jose, CA

May 2026 – Present

- Built a LangGraph-based multi-agent system to automatically analyze performance trace files using Trace Processor, identify performance bottlenecks and root causes, and localize issues within local code repositories.
- Developed an optimization planning and execution agent that translated profiling reports into actionable optimization strategies, performed code-level modifications, and validated improvements through profiling feedback.

Convolutional Layer CUDA Kernel Optimization

CUDA Developer

Champaign, IL

Jan 2026 – May 2026

- Implemented and optimized Convolutional Layer as matrix multiplication CUDA kernel, and using Tensor Core, shared memory tiling, constant memory caching, loop unrolling, CUDA streams, etc.
- Utilized Nsight Compute and Nsight Systems to analyze occupancy, memory access patterns, and Tensor Core utilization, iteratively optimizing GPU performance.

Job Application Management Web App

Backend Engineer

Champaign, IL

Sep 2025 – Dec 2025

- Developed a full-stack web application in a team that helps users track job applications.
- Built RESTful APIs in Django with well-structured data models and serializers to support scalable CRUD operations across multiple resources.
- Modeled relational database schemas using SQLite, enforcing data integrity through foreign keys and constraints.

Tiny-Chat

Programmer & Assistant

Virtual

Aug 2025 – Sep 2025

- Coded for a multi-agent conversational environment framework, which allows large language models to simulate social interactions such as negotiation, persuasion, and collaboration.
- Implemented data loading and random sampling of agent profiles from databases (Local/HuggingFace datasets), and be able to save conversation to JSON file.

Unix-like Kernel

Programmer & designer

Virtual

May 2025 – Aug 2025

- Built a Unix-like kernel from scratch using RISC-V and C in a 3-person team, implementing multitasking, file system, virtual memory, etc.
- Deployed a 2-way associative cache, enabled process management and cross-process communication via a pipe. Wrote drivers for different devices such as block device and entropy device.

Minerva Game Studio

Game Programmer

Madison, WI

Sep 2023 – Mar 2025

- Coded for a student-run studio to develop a stand-alone roguelike game using Unity Engine, whose demo is released on steam.
- Designed and built skills (ability of game characters) and assisted implementing language localization to enhance gameplay and improve player adaptability.

Skills

Languages: Mandarin (Native), English (Fluent), Japanese (Functional)

Technical: C/C++, Python, Java, C#, Git, Unity, LaTeX, PyTorch, Html, Javascript, CSS, CUDA